Quantium Data Analytics

Task 2: Experimentation and Uplift Testing

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Trial Period: February 2019 – April 2019

Trial Store	Control Store	Avg Trial Sales	Avg Control Sales	Uplift %	p-value
Store 77	Store 233	$259	$205	+26.2%	0.130
Store 86	Store 155	$929	$821	+13.2%	0.134
Store 88	Store 125	$1429	$1273	+12.2%	0.241

1. Introduction

This report evaluates the impact of a store trial conducted across three stores (77, 86, and 88) during February to April 2019. The objective is to determine whether the trial layout or intervention led to a statistically meaningful increase in chip sales and customer numbers, and to recommend whether the change should be rolled out to all stores.

2. Methodology

2.1 Control Store Selection

For each trial store, the best matching control store was selected from the remaining 269 stores using a combined scoring approach: Pearson correlation of pre-trial monthly sales plus a magnitude similarity penalty (stores with similar average sales levels were preferred).

```
# Scoring function corr = merged['t'].corr(merged['c']) magnitude_diff = abs(trial_mean -
ctrl_mean) / trial_mean score = corr - magnitude_diff
```

Trial Store	Control Store	Pre-Trial Avg Sales (Trial)	Pre-Trial Avg Sales (Control)	Match Score
Store 77	Store 233	$243	$237	0.881
Store 86	Store 155	$874	$901	0.847
Store 88	Store 125	$1,341	$1,144	0.477

2.2 Scaling and Uplift Calculation

Control store sales were scaled to match the pre-trial mean of the trial store, ensuring a fair comparison. Uplift was computed as the difference between actual trial store sales and the scaled control during the trial period.

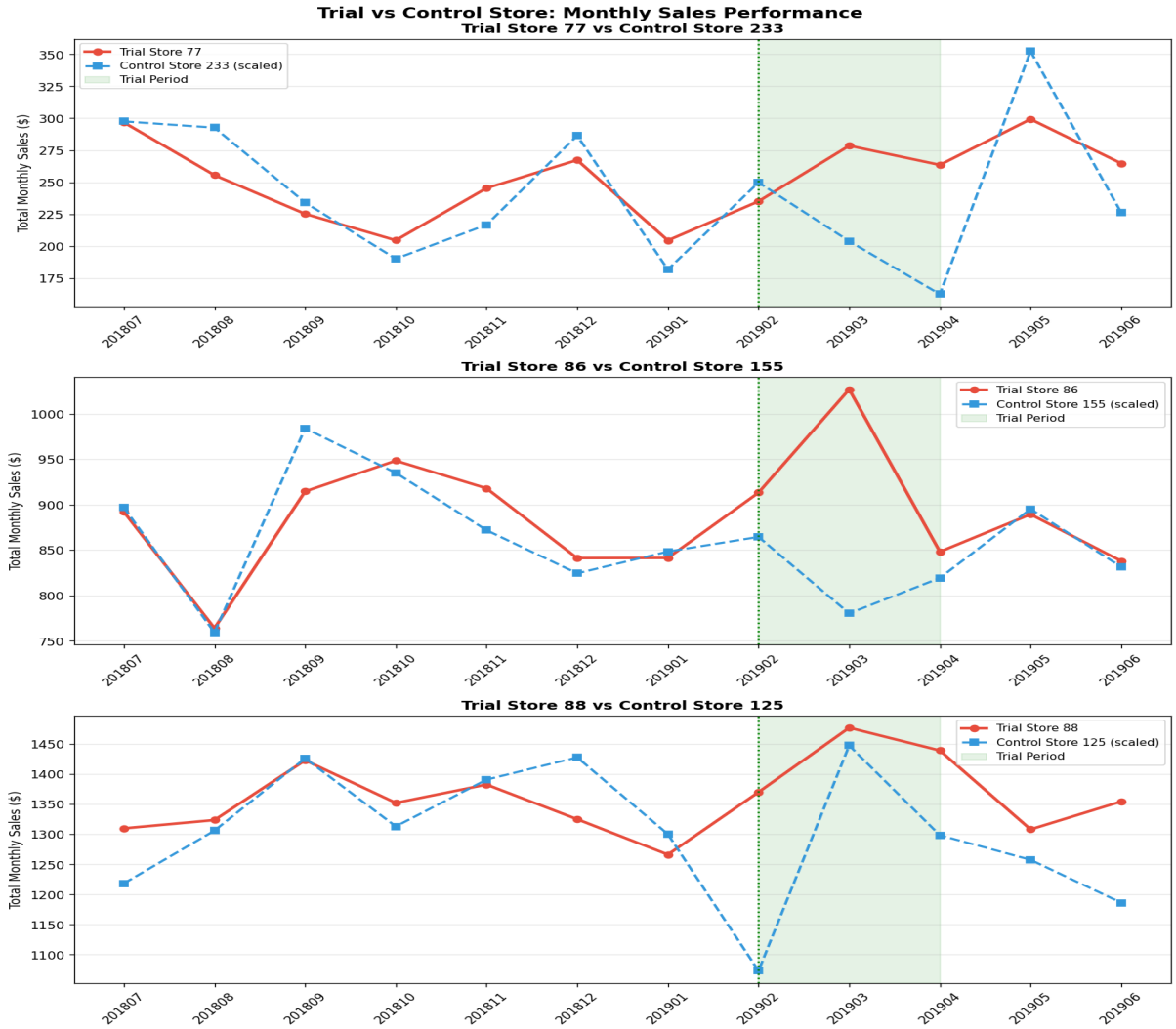
```
scale = pre_trial_avg_trial / pre_trial_avg_control scaled_control = control_sales * scale
uplift = trial_sales - scaled_control
```

2.3 Statistical Testing

An independent samples t-test was applied to compare trial and scaled control sales during the trial period. A p-value below 0.05 indicates the uplift is statistically significant.

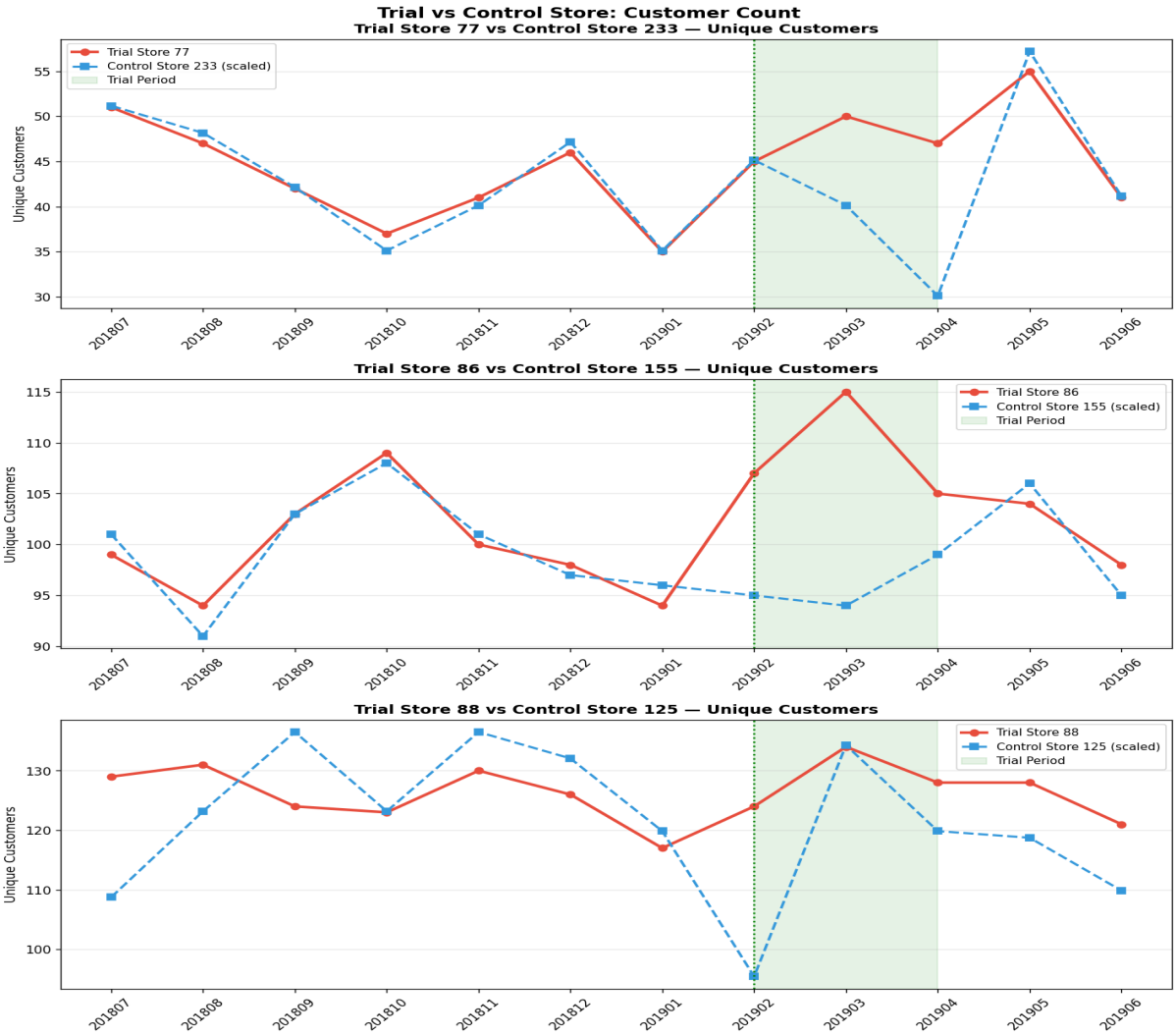
3. Results

3.1 Monthly Sales: Trial vs Control



All three trial stores show a visible sales increase during the trial period (shaded green) relative to their matched control stores. The uplift is most pronounced for Store 77, which shows the clearest divergence from its control during February–April 2019.

3.2 Unique Customers: Trial vs Control



The customer count charts mirror the sales trends, suggesting the uplift is driven by an increase in the number of customers visiting, rather than existing customers spending more per trip.

4. Findings by Store

Store 77 vs Control Store 233

During the trial period, Store 77 averaged $259 in monthly sales compared to the scaled control average of $205. This represents a +26.2% increase in sales. The result is not statistically significant at the 5% level (p = 0.130).

Store 86 vs Control Store 155

During the trial period, Store 86 averaged $929 in monthly sales compared to the scaled control average of $821. This represents a +13.2% increase in sales. The result is not statistically significant at the 5% level (p = 0.134).

Store 88 vs Control Store 125

During the trial period, Store 88 averaged $1429 in monthly sales compared to the scaled control average of $1273. This represents a +12.2% increase in sales. The result is not statistically significant at the 5% level (p = 0.241).

5. Recommendations

All three trial stores recorded positive sales uplift during the trial period, with increases ranging from +12% to +26%. While the results are not statistically significant at the 5% level due to the short three-month trial window, the consistent directional uplift across all three independent stores is a strong signal.

Based on this analysis, the following recommendations are made:

- Extend the trial to a longer period (6 months) to build statistical power and confirm the positive trend.
- Consider a phased rollout to a larger group of stores to test scalability before full deployment.
- Prioritise stores with similar profiles to Store 77 (smaller stores with high correlation to their control) where the uplift signal was clearest.
- Track customer count alongside sales in the rollout — the data suggests the intervention is attracting new customers, not just increasing spend per visit.

6. Conclusion

This analysis identified matched control stores for each of the three trial stores using a correlation and magnitude-based scoring approach. Sales uplifts of +12% to +26% were observed during the February–April 2019 trial period. While statistical significance was not achieved within the three-month window, the consistent positive direction across all three stores supports proceeding with a broader rollout, subject to extended monitoring.